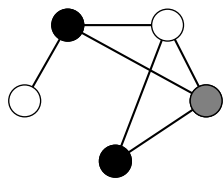


MAT 302: Discrete Mathematical Structures

Augsburg University – Fall 2024, §A

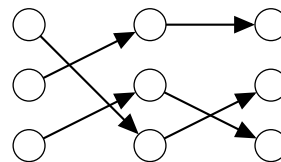


$$1+2+3+4+5+6+\dots$$

$$1, 1, 2, 3, 5, 8, 13, \dots$$

$$\{0, 1, 4, 9, \dots\}$$

$$\begin{array}{ccccccc} & & & & 1 & & \\ & & & & 1 & & 1 \\ & & & 1 & 2 & & 1 \\ & & 1 & 3 & 3 & & 1 \\ 1 & & 4 & 6 & 4 & & 1 \end{array}$$



General Information

Class meets MWF11:10 AM-12:20 PM in Hagfors 152. Our last quizzes are Wed Dec 11 from noon-2:00 PM which is our assigned final exam time. Classes meet in person but might occasionally switch to Zoom.

Professor Dr. Suzanne Dorée, Hagfors 185, doree@augsb.org You are welcome to call me Professor Dorée, Professor Su, or just Professor.

Office/help hours No appointment is necessary to come get help during my office hours – just drop in! My office hours are 1/2 hour long and meet in-person in the study area across from my office.

Mon 8-8:30 AM, 1-1:30 PM, 2-2:30 PM, 3-3:30 PM

Wed 1-1:30 PM, 2-2:30 PM, 3-3:30 PM

Fri 1-1:30 PM

I appreciate your sticking to the official times. I will give you advance notice if I have to change my office hours. If I am not available try working with a classmate, or e-mail me.

Textbook & technology

- Our free textbook is by Professor Su Dorée. Each section we cover is posted to Moodle.
- You will need regular access to the Internet to read email and to access resources on Moodle. If you do not have reliable Internet at home, contact the Info/Tech Desk at Lindell Library for help. (They can also connect you with a laptop or noise-cancelling headphones to help you work from home.)
- You will need regular access to a device that scans written work; any cellphone with a camera has this functionality. The on-campus printers also scan, but not as well. Alternatively, you can use a writable electronic surface (such as a tablet).
- Many days it will be useful to have an Internet-capable device in class that you can use to see the activities in the textbook (a cellphone, tablet, or laptop) but you will be working with a classmate, so often it is possible to share. Alternatively, you may print out each section before class. Note that you can print free on campus and if you run out of prints you can ask for more free prints.

Prerequisites MAT 145 *Calculus 1* with a grade of C- or higher (or P). If you received B- or lower in the prerequisite or any college mathematics course or if you are worried about taking a math class, please meet with me early in the semester and I will help you plan for success.

Standards met This course, with a grade of C- (or P) or higher

- Is required for the Mathematics Majors (BA and BS degrees)
- Required for the Computer Science Major (required for BS degree, option for BA degree).
- Option for Augsburg's Education Licensure programs in middle school (grades 5-8) and secondary (grades 5-12) mathematics. Details on the Minnesota Board of Teaching Standards met in full or in part by this course are available on request.
- Counts towards the Computer Science Minor, Mathematics Minor, and Data Science Major (in pair with MAT 350 Graph Theory)

Students who have completed MAT 271, MAT 302, or their transfer equivalents may not register for credit. If you have taken a full-semester discrete mathematics course at Augsburg or another institution, please meet with me right away.

Course topics

- *Enumerative combinatorics*: organized listing, counting techniques, the arithmetic triangle, and combinatorial proof.
- *Graph theory*: terminology, adjacency matrices, connectivity, isomorphism, invariants, coloring, complement, and trees.
- *Discrete structures*: strings, sequences, sums, and products.
- *Induction and recursion*: recursively-defined sequences, the Fibonacci sequence, modeling with recurrence relations, and proof by mathematical induction.
- *Number theory*: division of integers, primes, modular arithmetic, congruence mod n , greatest common divisor, the Euclidean algorithm, and alternate base representations (including binary and hexadecimal).
- *Formal logic*: logical connectives, quantifiers, negations, logical equivalences and inferences, and basic proof structures (contraposition, contradiction, cases).
- *Set theory*: subsets, set operations, Cartesian products, relations and equivalence relations with an emphasis on finite and discrete sets.
- *Functions*: function notation and visual representations with an emphasis on finite and discrete functions, one-to-one, onto, composition, inverses, and permutations.
- *Discrete algebra*: factorials, exponentials, log base 2, polynomials, and fractions.

Student learning objectives

- Learn basic concepts of discrete mathematics – enumerative combinatorics, graph theory, discrete structures, induction and recursion, and number theory.
- Be introduced to the foundations of theoretical mathematics and computer science – formal logic, set theory, and functions.
- Develop a discrete perspective and toolkit of techniques to solve problems that typically arise in MIS or computer science. These techniques include: recognizing patterns, organizing lists, formulating and testing conjectures, constructing counterexamples, explaining calculations and reasoning, reading definitions, arguing logically, and writing proofs.
- See famous problems and have fun solving mathematical puzzles.
- Work on the following dispositions from the ACM 2020 computer science curriculum recommendations¹
 - **Collaborative**: Team player, willing to work with others.
 - **Inventive**: Exploratory. Look beyond simple solutions.
 - **Meticulous**: Attentive to detail; thoroughness, accurate.

How to learn in this class (or tips for getting an A)

Being in class to have this opportunity to learn with me and classmates is very important. We will work on learning the new material through mini-lectures and working in pairs on activities. I will introduce new ideas, summarize what we are learning in the activities, discuss important course policies, and remind you of important deadlines. You will not want to miss class! I understand that at times you might occasionally have a serious situation that leads to your missing class. Please try to avoid missing class in less serious situations. We will use all of class time so please arrive on time and stay for the full session.

Building a supportive classroom where you feel welcome is a priority for me and I ask your help making sure all students are welcome and included. I have various course policies to support this work. In particular, during lecture time I do not allow voluntary calling out of answers, asking questions, or unsolicited comments. When you have questions during lecture (which will happen), you can quietly ask your neighbor or write down your

¹Available at <https://www.acm.org/binaries/content/assets/education/curricula-recommendations/cc2020.pdf>

questions for during activities or office hours. I will assign seats and weekly partners so you have an opportunity to work with many different classmates. I encourage you to embrace these practices, even if they seem a bit new to you, as I believe they will help you succeed.

The homework is posted to Moodle and submitted on Moodle. Homework problems often ask for you to explain and show your work, which is also good practice for the quizzes (and life). While many problems are relatively straightforward, I like to include more challenging problems too, so do not be surprised if you get stuck or are not able to complete every problem fully. That is okay and you should just turn in whatever you can do. Homework is typically due MWF 11:59 PM, although it is accepted through 11:59 AM (just before noon) the following day without penalty and until the start of our quiz (subject to late penalties). Make it a goal to not have any 0s on any homework.

There are two practices that will save you time and increase how well you learn the material. First, get in the habit of starting to starting the homework right away after class, at least to fill in whatever you can from what we did in class. Second, try to finish the assignment before our next class session, even though it is not due until later that day. That way you can meet with classmates to compare work and come see me during office hours (or email me) to get your questions answered well before the last minute.

You are welcome to work with classmates on assignments or to ask me questions during office hours or via email. If you work with a classmate, write up the final solution yourself to make sure you understand it. Please do not copy from classmates, look for solutions online, or otherwise compromise the integrity of your work. Plus, those strategies might get an assignment “done” but they will not help you learn the material.

There are no exams and no final exam in this course. Instead, we have quizzes every week or so. Most quizzes are part on Moodle and part in-class. On the Moodle part, you are welcome to use your class notes, assignments, and the textbook and you may discuss the concepts with classmates (but do not let a classmate answer the question for you), but it would be wise to try it on your own first to see how well you know the material (and what is left to study). The Moodle part can be repeated until you get 100%. The in-class part is closed book and individual – time to show me what you have learned. No reference material, computational technology, or collaboration is allowed on the in-class part of quizzes, unless we have alternative accommodations in place.

Study for the quizzes by reviewing class notes and your completed assignments. Be sure to review your “feedback files” on Moodle under each written homework and to meet with me or a classmate to discuss anything you do not yet understand. Be sure to understand how the quizzes are used in your course grade – you cannot get a high grade with low quiz grades.

How to upload your homework solutions

You may write your solutions by hand on paper or on a writable electronic surface (such as a tablet). Start by writing the section and your full name on the top of the first page. Organize your solutions are in order by problem number, number each problem and each part of the problem, leave several lines of blank space after each part of each problem and in between problems (so the homework grader can give you feedback), and to write clearly using black/blue ink (or dark pencil). If you are writing on paper, scan (not photo) the pages into a single PDF when you are done. If you are writing on an electronic surface export/print to PDF when you are done. Be sure to look at your scan/upload to make sure it is a readable and not gigantic. Then upload to Moodle. Moodle accepts 1 PDF file per homework. I do not accept homework via email.

How your homework is graded

First, each problem is marked using this scale:

- 5** Complete, correct, and clearly written solution.
- 4** Nearly complete, correct, and clearly written solution.
- 3** Some correct, relevant progress toward a solution OR correct answer with no work.
- 2** Work shows understanding of the problem but little or no progress toward a solution.
- 1** Attempted but does not show understanding of the problem.

0 Not attempted

Then the homework set is scored using the following formula: $\left\lceil \frac{\text{Total of Problem Points}}{\text{Number of Problems}} \right\rceil$

Each homework is graded out of 5 points. Because we are rounding up, you can earn 5 points (100%) but still have some problems that were not correct. Be sure to look back at every problem and the feedback from the grader when studying for the quizzes.

Late homework is accepted up to the start of class the day of the quiz on that topic. Late work is not graded for feedback, but you can earn up to 2 points (out of 5). It is usually better to turn in a half-finished homework on time than to submit homework late.

At times I will offer extra credit work. Some extra credit work corresponds to activities called “puzzles”. You must have been in class working on the puzzle to be eligible to submit extra credit work on those puzzles. If you miss class, you are not eligible to submit extra credit work on the puzzle you missed, but do not worry – there’s a lot of extra credit opportunities available!

How your activities are graded

If you are in class from start to finish and paying attention, you get 5 points. If you arrive late, step out during class, leave early you usually get 4 points. If you miss class or are not participating you get 0 points. You can earn back points up to a max of 4 points by writing up the activities and giving it to me in person or via email within a week of the class. You are welcome to get help from me or a classmate in completing miss activities (but please do not just copy). Don’t forget to submit activities for any day you miss.

How your quizzes are graded

Each quiz is worth 20 points. During our assigned final exam time we have two quizzes. Often the Moodle part is worth 5 points calculated on the following scale:

min % on Moodle part	100%	90%	80%	70%	60%	50%
quiz points	6 (yes!)	5	4	3	2	1

and the in-class part is worth 15 points.

What I expect of you (and course policies you should know)

- Check your Augsburg email account regularly and access our Moodle site as needed.
- Attend and participate in every class. Be on time and prepared for every class. Arrange to stay for the entire class period. If you must come late or leave early on special occasion, please let me know in advance. Avoid scheduling appointments that overlap class time. If you have an occasional excused absence, send me an email as soon as possible so that I know you are okay.
- Pay attention during class. The only thing you should be reading, viewing, writing, or discussing during class is the current material. You may use electronic devices that lay flat on the table (e.g. cellphone or tablet) for taking notes including photographs of the whiteboards, calculations, or approved accommodations, but not for other purposes during class. Laptops are usually not allowed for note-taking (unless they lay flat or you have a documented disability-based need.) When we use cellphones, tablets, or laptops to access activities during class, that is the only thing you should be looking at on those devices.
- Do all of the assigned work. Plan on spending 6-9 hours a week outside of class time working on the course. Try scheduling at least 2 hours of work time (perhaps 1 hour + 2 hour) between every pair of class sessions. Your study time should include reviewing class notes and activities, reading the textbook, working on homework, taking the Moodle part of quizzes, and studying for the in-class part of quizzes.
- Submit work when due. You may submit work for as long as the link on Moodle allows, subject to late penalties. I reserve the right to limit how many and which assignments are eligible for late submission.

- Get help when you need it. I know it is hard to ask for help sometimes, but I really enjoy helping students to learn so please reach out to me for help as you need it. You would be wise to arrange to meet with a classmate outside of class to compare work as well. Do not wait until the last minute to get help or meet with classmates. If you did not do well in previous college mathematics courses, or it has been a few years since you took those courses, you should make every effort to arrange your schedule to be able to attend office hours.
- Take quizzes, including those during our assigned final exam time, when scheduled. If you have a conflict with a date, discuss the options with me well in advance. If you miss a weekly quiz without having made prior arrangements, get in touch with me as soon as possible to ask if you can take a make-up quiz. While I often allow make-up quizzes, understand that a missed quiz may result in 0 points. I reserve the right to limit the number of make-up quizzes and to designate the time period during which make-up quizzes are offered. I also reserve the right to require follow-up conversation with me during office hours to check on your understanding of the material for any quiz.
- Work with others in the class when asked to do so. Work alone when asked to do so. Follow all academic honesty rules. To be clear, you are encouraged to share ideas, check answers, and compare solutions or explanations with classmates, but any work you submit must be "your own work" as defined by the Augsburg University Academic Honesty Policy. Direct copying is forbidden. You may not get solutions from additional texts or resources or from people not currently in the class without my written permission. In particular, you are not allowed to work with tutors or other professors without my written permission. The consequence of a violation of this policy is anywhere from a 0% on the work to an F in the course. I am also required to notify the Provost's office who may impose further consequences. Please protect your reputation and uphold your integrity by committing to being academically honest throughout this course. If you ever unsure if something is allowed, please ask me.
- Check Moodle gradebook regularly and email me if you have any questions about your grades. If you do late work and don't see a grade entered in Moodle after a week, feel free to email me.
- Keep me up-to-date on how the class is going for you. Be sure to come in and talk with me if I email you or send an Academic Alert so that we can get you back on track.
- Let me know of any special needs or concerns you have. If you need accommodations because of a disability, please contact the CLASS Office (612) 330-1053 or class@augsborg.edu.
- Creating a classroom environment that welcomes and values all students is very important to me. I ask your assistance in helping support one another by following classroom rules. If you feel unwelcome or unvalued at any time, or have any suggestions for improvement, please let me know so I can try to help. Please do not be shy about telling me what is or is not working for you.
- Complete any in-class and online course evaluation forms.
- Understand and comply with university-wide measures to reduce COVID-19 and influenza transmission. I greatly appreciate it if you stay away or wear a mask if you are sick and might be contagious, even if it's "just" a cold. You are welcome to wear a mask at any time.
- As with all Augsburg classes, you should understand and comply with all University policies described in the University catalog and the Student Guide: <http://inside.augsburg.edu/studentaffairs/studentguide/>

How I calculate your course grade

First, I divide the total points you earned for Activities and Homework by the total number of points available to get a percentage grade on Activities and Homework.

Next, I divide the total points you earned on Quizzes by the total number of points available to get a percentage grade on Quizzes. Then, I compute the weighted average

$$\begin{array}{c}
 65\% \text{ of percentage grade on Activities and Homework } (\approx 200 \text{ points each for } \approx 400 \text{ points total}) \\
 + \\
 35\% \text{ of percentage grade on Quizzes } (\approx 300 \text{ points})
 \end{array}$$

to get your overall percentage. The overall percentage is approximately the percentage of course points you earn – so think of each point on anything as being worth one point.

Last, I convert your percentage grade to a letter grade using the following specifications.

- A requires at least 94% overall and at least 84% on quizzes.
- A- requires at least 92% overall and at least 82% on quizzes.
- B+ requires at least 89% overall and at least 79% on quizzes.
- B requires at least 84% overall and at least 74% on quizzes.
- B- requires at least 82% overall and at least 72% on quizzes.
- C+ requires at least 79% overall and at least 69% on quizzes.
- C requires at least 74% overall and at least 64% on quizzes.
- C- or P requires at least 72% overall and at least 62% on quizzes. This grade is the minimum required to count the course towards the mathematics major or minor, computer science major or minor, data science major, for Education programs, or as a prerequisite for any MAT or CSC course.
- D or LP requires at least 62% overall and at least 52% on quizzes. This grade is the minimum required to receive credit towards graduation for the course.
- Any other situation is graded F or N.

I reserve the right to lower your course grade if your attendance, punctuality, participation, preparation, or behavior falls well below expectations. If during the term you are not progressing towards a passing grade or if your attendance, participation, preparation, or behavior is falling well below expectations, I will usually send an Academic Alert.

I reserve the right to lower your course grade if you violate the Academic Honesty Policy in this course. If that happens, I will notify you in writing. Again, if you are ever in a situation where you are not sure if something is allowed, please ask!

You are welcome to take this course P/LP/N (Pass/LowPass/No Credit) but you should check with your academic advisor first because there is a limit to the number of P/N courses overall, towards a major, and towards a minor. Some students like this option because the grades of P, LP, and N do not count into your Grade Point Average (GPA). Pay attention to the posted deadline to change grading option (on the Augsburg Registrar's website).

For this course to serve as a prerequisite for MAT or CSC courses you must earn a C- or higher (or P). While a C- or higher (or P) satisfies prerequisites, any grade below a B usually means you will have significant challenges in advanced mathematics or computer science classes. So, if you're planning to take upper division coursework in mathematics or computer science, you should work towards a B or higher in this course to be adequately prepared to go on.

MAT 302 Calendar – Fall 2024, §A subject to change with notice.

Monday	Wednesday	Friday
	August 28	August 30
	1.1 Getting Started	1.2 Organized Listing
September 2 No classes: Labor Holiday	September 4 1.3 Counting with Steps and Cases	September 6 1.4 Counting Subsets
September 9 1.4 continued Quiz: 1.1-1.2	September 11 2.1 Integer Division	September 13 2.2 Modeling with Graphs
September 16 2.2 continued Quiz: 1.3-1.4	September 18 2.3 Functions and Function Notation	September 20 2.4 Functions on Discrete Objects
September 23 3.1 Logical Connectives, Quantifiers, and Negations Quiz 2.1-2.2	September 25 3.2 Conditional Statements	September 27 3.3 Number-theoretic Functions
September 30 3.4 Divisors and Primes Quiz 2.3-2.4	October 2 3.4 continued	October 4 4.1 Explicitly-defined Sequences
October 7 4.2 Sums and Products Quiz 3.1-3.2	October 9 4.3 The Arithmetic Triangle	October 11 4.3 continued & 4.4 Binary and Hexadecimal Representation
October 14 4.4 continued Quiz 3.3-3.4	October 16 5.1 Recursively-defined Sequences	October 18 No classes: Midterm break
October 21 5.2 Mathematical Induction Quiz 4.1-4.2	October 23 5.2 continued	October 25 5.3 Matrices
October 28 5.4 Sum and Product Formulas Quiz 4.3-4.4	October 30 6.1 Subsets and Set Operations	November 1 6.1 continued & 6.2 Cartesian Products, Vectors, and Relations
November 4 6.2 continued Quiz 5.1-5.2	November 6 6.3 Probability	November 8 6.4 Sophisticated Sums
November 11 7.1 Graph Theory Quiz 5.3-5.4	November 13 7.2 Graph Isomorphism	November 15 7.2 continued & 7.3 One-to-one and Onto
November 18 7.3 continued Quiz 6.1-6.2	November 20 7.4 Finite Groups	November 22 8.1 Properties of Relations and Congruence
November 25 8.1 continued Quiz 6.3-6.4	November 27 8.2 Equivalence Relations	November 29 No classes: Thanksgiving
December 2 8.3 Logical Equivalence Quiz 7.1-7.2	December 4 8.3 continued	December 6 Last day of class 8.4 What Does “Discrete” Mean?
December 9 No classes: Finals Week	December 11 (12-2 PM) Quiz 7.3-7.4 & Quiz 8.1-8.3	Subject to change with notice. Effective Aug 28